



Fraud detection analysis in financial Transactions

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Abstract: *Fraud in financial transaction is a crucial issue for banking industry. Fraud mostly occurs in cash-out and transfer transaction. The classification of imbalanced data remains a significant challenge in machine learning, particularly in fraud detection. To address the imbalance between majority and minority classes, techniques such as undersampling, SMOTE (Synthetic Minority Oversampling Technique), and random oversampling are employed to improve classification performance.*

Keywords: Imbalanced data, Transaction detection, Oversampling, Under-sampling, SMOTH, Machine learning

1. INTRODUCTION

Financial fraud poses a significant threat to businesses, banks, and consumers, leading to substantial economic losses worldwide. As digital transactions increase, traditional rule-based fraud detection systems struggle to identify sophisticated fraudulent activities. Machine learning offers a robust solution by analyzing large-scale transaction data and detecting anomalies that indicate potential fraud. In today's digital age, financial transactions are increasingly vulnerable to fraud. As more people rely on online payments and cashless systems.

This study explores multiple machine learning models for fraud detection. Given the highly imbalanced nature of fraud datasets, we employ class balancing techniques such as SMOTE, under-sampling, and oversampling to improve model training. A comparative analysis is conducted across models, including Logistic Regression, Decision Trees, Random Forest, evaluating their performance using precision, recall, F1-score, and AUC-ROC. The findings demonstrate the importance of robust feature engineering and class-balancing strategies in enhancing fraud detection accuracy.

This research contributes to the ongoing efforts in securing financial transactions by optimizing fraud detection models and ensuring a balance between accuracy, speed, and minimal false positives.

2. RELATED WORK

Using various machine learning techniques such as supervised, semi-supervised, and unsupervised learning, researchers have created a number of models to automate

financial fraud detection systems. (Ileberi et al., 2021) proposed a machine learning approach for credit card fraud detection using imbalanced real-world datasets. To address class imbalance, they applied the synthetic minority oversampling technique (SMOTE). Various models, including SVM, linear regression, random forest, XGBoost, decision tree, and extra tree, were evaluated. To enhance accuracy, these models were combined with adaptive boosting (AdaBoost). Performance was assessed using metrics such as Matthews correlation coefficient (MCC), AUC, recall, and precision. (Salekshahrezaee et al., 2023) explored the impact of feature extraction and data sampling on credit card fraud detection using a dataset and four ensemble classifiers. They evaluated random under sampling (RUS), SMOTE, and SMOTE Tomek for data sampling, and PCA, convolutional autoencoder (CAE), and RUS for feature extraction. The best performance in fraud detection was achieved by combining RUS with CAE. (Ahmad et al., 2023) developed a method for handling imbalanced data by combining under-sampling and fuzzy C-means clustering to select similar fraud and normal instances with matching attributes. This approach aims to preserve data features while enhancing accuracy and performance across different machine learning methods.

3. METHODOLOGY

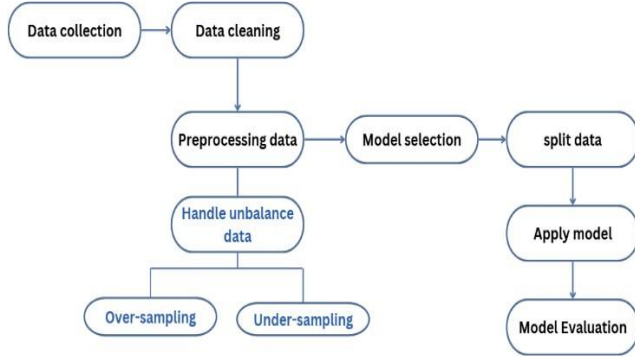


Fig.1. Methodology

3.1 Dataset

Our dataset collected from open sources website, Kaggle. The dataset's name "Fraud.csv" (*Fraudulent-Transactions-Data*, n.d.). The dataset consists of 6,362,620 transactions and contains two unbalance classes, 635,4407 are normal transaction and 8,213 are fraudulent. For the more detail, s about dataset has 11 attributes such as step, type, amount, oldbalanceOrig, newbalanceOrig, oldbalanceDest, newbalanceDest, isFraud, and isFlaggedFraud.

3.2 Data cleaning

We check the data there are any missing value or duplicated value or not, but our dataset not contains both of them.

3.3 EDA

Table 1 and table 2 below show about the unbalance classes (fraudulent and non-fraud) and type of transaction.

Table1. unbalance classes

Classes	Number	Percentage
Fraud	8,213	0.13%
Non-Fraud	635,4407	99.87%

Table 2. Type of fraud

Type	Fraud	Non-fraud
Cash out	4116	2,233,384
Transfer	4097	528,812
Cash in		1,399,284
Debit		41,432
Payment		2,151,495

Fraud detection analysis

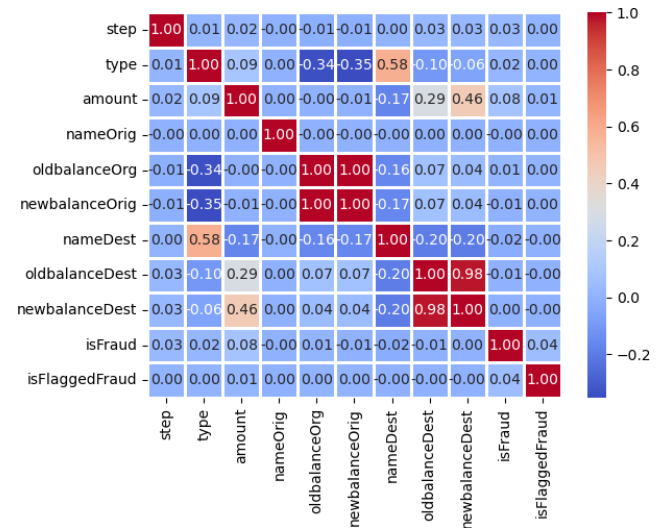
From dataset show that transaction that all transactions which contain 'amount'=0, are fraud transactions. It is obvious that if someone having null amount in their account, thus there is no sense to transfer money. In the table 2, it is clear that fraudulent agents only targeted "CASH_OUT" and "TRANSFER" type of trans. Because 100% fraud transactions are lie inside these two types.

3.4 Preprocessing data

A. Feature engineering

In this section, will explore how to select variable to be included in the model. We use correlation matrix to find the relationship between each variable. With correlation heatmap in the fig.2 below, it shows that the highest correlated is newbalanceOrig and oldbalanceOrig and higher correlated is newbalanceDest and oldbalanceDest.

Fig.2 . Correlation heatmap



We concern about multicollinearity, so we use Variance Inflation Factor (VIF). VIF is a statistical measurement that indicates the severity of multicollinearity. VIF method detection:

- VIF = 1: No multicollinearity
- VIF between 1 and 5: Moderate multicollinearity
- VIF > 5: High multicollinearity
- VIF > 10: Serious multicollinearity

By VIF, oldbalanceOrig and newbalanceOrig, oldbalanceDest and newbalanceDest are too high VIF. Thus combine these pairs of collinear attributes and drop the individual ones. So, new variables we get are:

$\text{Actual_amount_orig} = \text{oldbalanceOrg} - \text{newbalanceOrig}$
 $\text{Actual_amount_dest} = \text{odlbalanceDest} - \text{newbalanceDest}$
 $\text{TransactionPath} = \text{nameOrig} + \text{nameDest}$
 Therefore, our features: type, amount, Actual_amount_orig, Actual_amount_dest, TransactionPath and target variable is isFraud.

B. Handle unbalance classes

Resampling technique are a common set of strategies used to address data imbalance in machine learning. These techniques involve modifying the data by either increasing the number of minority class sample (oversampling) or reduce the number of majority class samples (under sampling). oversampling technique was the SMOTH method known as Synthetic Minority Oversampling Technique (Ashtiani & Raahemi, 2022). SMOTE generates synthetic instances for the minority class by interpolating between neighboring instances.

To handle this problem, we use two difference methods: under-sampling and oversampling. Because we want to compare the differences between them in model performance.

Under-sampling is a technique that reduce majority as minority class to balance dataset. Our dataset minority class (fraudulent) is 8213, so after under-sampling we got 8213 is normal transaction and 8213 is fraudulent transaction.

Oversampling is a technique that produce minority class the same as majority class. There are several techniques to increase minority class, but here we use only random oversampling and SMOTH. With random oversampling we got two classes and each contains 6354407. And SMOTH we got 12708814 in total.

3.5 Model selection

3.5.1 Random Forest

Random forest is an algorithm model for classification for machine learning. For classification, the output of the random forest is the class select by most tree. The algorithm combining multiple decision trees to improve accuracy and robustness, well-suited for handling imbalanced dataset in fraud detection. Random forest is another ensemble learning technique for regression, classification, and other which tend to achieve higher performance (Ashtiani & Raahemi, 2022).

3.5.2 Decision Tree

A straightforward model that splits data into decision-based rules, making it effective for identifying clear patterns in financial fraud (Quinlan, 1986).

3.5.3 Logistic Regression

A simple and interpretable method for binary classification, ideal for modeling the probability of fraud occurrence based on feature relationships (Hosmer et al., 2013). Moreover, the most Machine learning model used in developing financial misstatement detection is the logistic regression model (Ashtiani & Raahemi, 2022).

3.6 Split data

For training and test set, we use 80% of data for training set and 20% for test set.

3.7 Apply model

In this section we demonstrate applying machine learning models, Logistic Regression, Decision Tree and Random Forest into three differences balance dataset.

4. RESULTS AND DISCUSSION

4.1 Results

With under-sampling technique, model performed is Logistic Regression (LR) 89.96 % accuracy, Decision Tree (DT) 96.20% and Random Forest (RF) 97.02%. In detail see table below. (See Table 3)

Moreover, with SMOTH balance dataset model performance is Logistic Regression (LR) 93.02%, Decision Tree (DT) 99.38% and Random Forest (RF) 98.51%. And random oversampling Logistic Regression (LR) achieved 92.78%, Decision Tree (DT) 99.98%, and Random Forest (RF) 99.98%. (See Table 4 and 5)

Table 3. Result of using under-sampling technique

Models	Accuracy	Precision	Recall	F1-score
LR	0.89	0.91	0.90	0.90
DT	0.96	0.96	0.96	0.96
RF	0.97	0.97	0.97	0.97

Table 4. Result of using SMOTH technique

Models	Accuracy	Precision	Recall	F1-score
LR	0.93	0.93	0.93	0.92
DT	0.99	0.99	0.99	1.00
RF	0.98	0.99	0.99	0.99

Table 5. Result of random oversampling

Models	Accuracy	Precision	Recall	F1-score
LR	0.93	0.93	0.93	0.93
DT	0.99	1.00	1.00	1.00
RF	0.98	1.00	1.00	1.00

4.2 Discussion

The LR model performs best with random oversampling (93.02%) and slightly worse with SMOTE (92.78%) and under-sampling (89.96%). Tree-based models like DT and RF achieve near-perfect accuracy with SMOTE and random oversampling (99.98%). These models benefit from the increased minority class representation provided by SMOTE and random oversampling, improving their ability to detect fraudulent transactions. Under-sampling, which reduces the majority class, leads to lower performance, particularly for LR. Overall, SMOTE and random oversampling yield the best results, especially for tree-based models, when handling imbalanced datasets. Careful attention should be paid to the potential overfitting risk with random oversampling.

5. CONCLUSIONS

The study demonstrated that handling data imbalance using SMOTE greatly improves model performance in fraud detection. Among the models, Random Forest and Decision Tree emerged as the most effective due to their ability to capture complex patterns and maintain high performance across classes. Logistic Regression, while interpretable, was less effective in scenarios with imbalanced data. Future work could explore further optimization of Random Forest parameters and the integration of hybrid sampling techniques for even better results.

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